AutoDoomscroller: An On-Device Agent That Learns Which Short Videos You Would Skip

Project Proposal

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ABSTRACT

Short-video feeds are ranked to keep people watching, yet in six days of our own logs 38% of the non-ad reels were swiped away within two seconds. We propose AutoDoomscroller, an Android agent that learns from one person's scrolling which Instagram Reels and YouTube Shorts they would skip, then skips them, lets them scroll and like with gestures, and shows them what they watch. Everything runs on the phone. A prototype already records both apps, and a first model reaches a skip AUC of 0.63.

1 MOTIVATION

Scrolling a short-video feed means making the same small decision hundreds of times a day: watch this, or swipe it away. In six days our prototype logged 1,833 reels. Of the non-ad reels, 38% were left within two seconds, half of those within about one second (Fig. 1). The feed itself is chosen by the platform's recommender, which is built to maximise watch time [1, 11], and users of such feeds report a weakened sense of agency [6].

We ask whether the phone can learn those quick judgements from the user's own behaviour and act on them: skip what the user would skip anyway, let them move through the feed without touching it, and show them where their time goes. The intended users are people who watch short videos every day and want less of that time spent on reels they don't care about, including when their hands are busy; in our logs that is about 100 quick swipes a day. Unlike the platform's recommender, the agent works for the user and keeps their data on the phone.

The project covers the course's core topics: mobile sensing (accessibility events, screen capture, camera), on-device inference under latency and energy limits, and model optimization (quantization, pruning) measured before and after.

2 LITERATURE REVIEW

Platforms predict engagement from the logs of millions of users: YouTube ranks videos by expected watch time [1], Kuaishou corrects watch-time prediction for video length [11], and dwell time is a well-established signal of preference [10]. EdgeRec moves part of the ranking onto the phone to react to the user's latest actions [2], but still for the platform's goals. We instead predict one person's skips on their own phone, from a few hundred examples and without access to the platform.

Digital self-control tools mostly block or limit use [7], and Lukoff et al. found that autoplay and recommendations undermine users' sense of agency [6]. Our agent does not block anything; it acts on the user's own past choices and makes them visible.

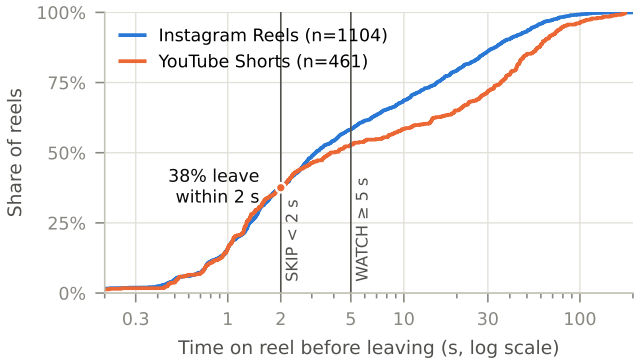


Figure 1: Time spent on each non-ad reel before swiping away (six days, one user). In both apps 38% of reels are left within 2 s.

For hands-free control, EyePhone showed early on that a phone can be operated with eye blinks [8], and MediaPipe now runs face and hand landmark models in real time on phones [5]. We use such gestures to move through a feed, and each gesture also becomes a training label. For the models we use MobileNetV3 on frames [3] and multilingual E5 on text [9], and compress them with integer quantization [4] and pruning.

3 PROPOSED DESIGN

Sense (Fig. 2). An accessibility service reads the screen of Instagram Reels and YouTube Shorts a few times per second to know which reel is showing, how long it stayed, and whether the user liked, saved or commented on it or watched it to the end. Once a reel has settled after the swipe, the service takes one screenshot cropped to the reel; in hands-free mode the front camera tracks the face and hands.

Label. A reel counts as SKIP if the user left within 2 s, and as WATCH if they interacted with it, watched it to the end, or stayed at least 5 s. Everything in between is left out. Likes alone would not work as a target: there were only 6 in 1,833 reels.

Learn. The model may only use what is known when a reel appears: the first frame, the text on it (read with on-device OCR such as ML Kit), the caption, and the creator's history. For each we look up the most similar reels the user has already seen and use

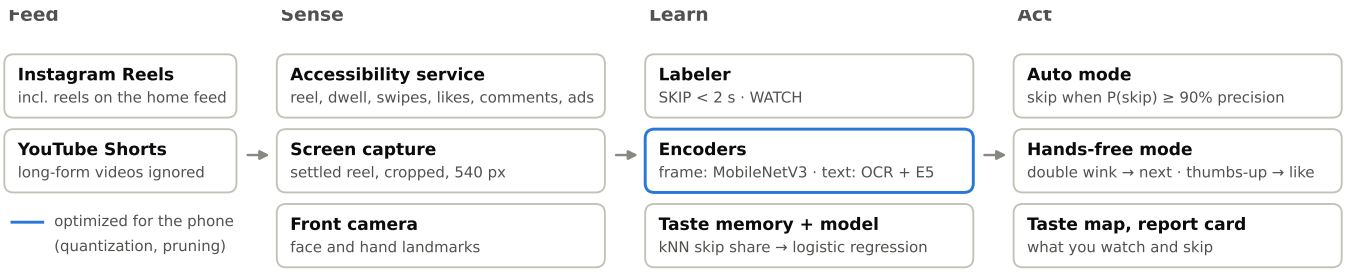


Figure 2: System overview. Everything runs on the phone.

Table 1: Skip AUC on 933 labeled reels, each day predicted from the days before it (mean, 95% CI).

Inputs	Skip AUC
Creator history	0.63 (0.58–0.68)
First frame (MobileNetV3)	0.56 (0.52–0.61)
Text on the frame (OCR)	0.58 (0.53–0.62)
Creator + frame + caption	0.62 (0.57–0.67)
+ session context	0.56 (0.52–0.61)

the share of skips among them as a score, so the model improves as the memory grows, without retraining the encoders. A logistic regression combines the scores into $P(\text{skip})$.

Act. Auto mode swipes when $P(\text{skip})$ is high enough to be right 90% of the time; if the user goes back to an auto-skipped reel, that counts as a correction. Hands-free mode maps a double wink to "next" and a thumbs-up to "like". A taste map and a per-session report card show the user's habits.

4 PROTOTYPE AND IMPLEMENTATION PLAN

The Android app (Kotlin, tested on a Galaxy S24) already tracks both apps, detects and skips ads, and exports its logs for offline analysis in Python, where we compute embeddings, run OCR and evaluate models on days they were not trained on.

Preliminary results. Table 1 shows the first results. The creator's history alone reaches an AUC of 0.63; adding the frame and caption does not help yet. The most promising content signal is the text on the frame: on reels where a sharper 540 px frame is available, its AUC rises from 0.46 to 0.63, so the app now captures that frame first. Session information (time of day, scrolling pace) hurts, as the user's mood shifts between sessions.

Tools and data. On the phone: LiteRT for the encoders, ML Kit for OCR, MediaPipe Face Landmarker and Gesture Recognizer for hands-free mode; offline: Python with scikit-learn and the TensorFlow Model Optimization Toolkit. Data comes from the author's own phone, about 300 reels a day, labeled automatically; no manual annotation is needed.

Steps. Nov. 1–7: proposal; collect data with the new capture. Nov. 8–14: hands-free mode. Nov. 15–21: better model; run it on the phone. Nov. 22–28: compression study; Auto mode. Nov. 29–Dec. 5: measurements, visualization, final report.

Team. One person: the author builds the app and models and runs the evaluation.

5 EVALUATION PLAN

- *Skip prediction:* AUC on days not used for training; goal ≥ 0.70.
- *Auto mode, live:* precision (from a random 10% of predicted skips that are let through) and the undo rate; goal ≥ 90% precision. Also the share of the user's quick skips it takes over.
- *Compression:* model size, latency per reel on the phone and AUC for FP32, FP16, INT8 and pruned encoders; goal: a decision within 300 ms of a reel's arrival, losing at most 0.01 AUC.
- *Hands-free and energy:* gesture accuracy, false triggers per 10 minutes, and battery drain per hour of scrolling in each mode.

6 POTENTIAL CHALLENGES AND SOLUTIONS

- *Weak, mood-dependent signal.* Read on-screen text from the sharper first frame, set the threshold per session, and keep updating the memory online.
- *Time and battery.* Half of the quick skips happen within about a second. One cropped screenshot is taken as soon as the reel settles, OCR runs on that frame only, the encoders run in INT8, and the camera runs only in hands-free mode at a low frame rate.
- *False gestures.* Natural blinks use both eyes and are short, so "next" needs a double wink of one eye and "like" a thumbs-up held for half a second.
- *Measuring an agent that acts.* Auto-skipped reels lose their label, so a random 10% of predicted skips are let through to measure precision.
- *App updates.* Each app has a small adapter keyed on view IDs; screen dumps make it quick to fix.

7 EXPECTED OUTCOMES

An Android app with Learning, Auto and hands-free modes running entirely on the phone; a personal skip model; a before-and-after comparison of the compressed encoders; and a taste map and report card showing how the user spends their scrolling time.

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